

Course Welcome

Introduction



Welcome to AWS Partner: Agentic AI Essentials. This course explores agentic AI solutions, advancing beyond generative AI fundamentals. Students will learn how agentic AI represents the convergence of distributed systems expertise with modern AI reasoning capabilities, building directly on the generative AI foundation they already understand.

Course learning objectives



This course consists of multiple modules. On completion of all modules, you will be able to do the following:

- Differentiate agentic AI from the generative AI capabilities learned in the prerequisite
- Articulate real-world applications where agentic AI transforms business processes
- Articulate the business value proposition for autonomous AI agents
- Identify customer scenarios where agentic AI provides superior outcomes
- Describe AWS's comprehensive agentic AI stack and competitive advantages
- Explain agent limitations, multi-agent coordination, and reliability considerations

AI agents or agentic AI: what's the difference?



AI agents are individual software programs designed to perform specific tasks within defined parameters. They operate much like specialized workers, focusing on singular purposes such as chatbot interactions, making recommendations, or executing automated trades. While these agents can sense their environment and take predetermined actions, their autonomy and decision-making capabilities remain relatively limited. They excel at their designated tasks but typically operate in isolation, without deeper understanding of broader contexts or the ability to significantly adapt their behavior beyond their programming.

In contrast, **agentic AI** represents a more sophisticated evolution in artificial intelligence, demonstrating true agency and autonomous behavior that goes beyond simple task execution. These systems exhibit goal-directed behavior coupled with complex reasoning capabilities, allowing them to understand and operate within broader contexts while making independent decisions. **Agentic AI can coordinate with other systems and agents**, modify its behavior based on experience, and work within complex environments to achieve objectives. Think of autonomous business process orchestrators or self-managing cloud systems that can understand the broader implications of their actions and adjust their strategies accordingly.

The fundamental difference lies in their capabilities and scope of operation. While AI agents function like specialized tools with specific purposes, agentic AI operates more like a capable manager who can understand complex situations, coordinate multiple activities, and make strategic decisions.

AWS Summit New York City 2025

Watch Swami Sivasubramanian, Vice President, Agentic AI, AWS, reveal groundbreaking agentic AI innovations and showcase latest AWS tools and services.

[Watch full Video.](#)



Agentic AI: welcome to the latest AWS partner ally

As trusted advisors, AWS Partners play a crucial role in helping customers navigate their journey from traditional applications to agentic architectures, ensuring successful adoption and value realization.

Agentic AI represents a new revenue stream for AWS Partners, enabling the development and deployment of intelligent, autonomous solutions that solve complex enterprise challenges. Partners can leverage AWS's comprehensive agent development stack - from foundation models to AgentCore - to build scalable, production-ready solutions for their customers faster.

The AWS Vision on Agentic AI

Introduction



AWS envisions billions of agents working alongside humans, augmenting their capabilities, accelerating innovation, and improving productivity across every industry. These agents will operate in consumer, enterprise, and industrial settings, orchestrating not just simple, single-step tasks but sophisticated, multi-step processes that require understanding context, making decisions, and taking actions.

AWS is committed to being the best place to build the world's most useful AI agents, empowering organizations to deploy reliable and secure agents at scale. AWS is focusing on making agentic AI vision accessible to every organization by combining rapid innovation with a strong foundation of security, reliability, and operational excellence.

This approach helps AWS Partners accelerate progress by building on proven principles while embracing new possibilities—creating systems that adapt as models evolve and new capabilities emerge, all built on AWS's secure, reliable agentic AI foundation and infrastructure.



What makes AWS uniquely suited for agentic AI is not just its cutting-edge technology but also its proven foundation of security, governance, scale, and reliability—critical qualities for deploying autonomous systems.

An unprecedented opportunity



Looking ahead to 2028, Gartner forecasts a dramatic rise in agentic AI adoption ([Gartner, Top 10 Strategic Technology Trends for 2025, published October 2024](#)).

Enterprise software applications incorporating agentic AI will surge from less than 1% to 33%. Additionally, while no work decisions are currently made autonomously by agentic AI, 15% will be by 2028. The transformation extends to generative AI services, where one-third of interactions will involve action models and autonomous agents for task completion. Adoption is moving quickly against that forecast. In the [2026 Gartner CIO and Technology Executive Survey](#), 17 percent of organizations had deployed AI agents, and more than 60 percent expected to do so within two years - the most aggressive adoption curve of any emerging technology in the survey.

of enterprise software apps will include agentic AI by 2028, up from less than 1% in 2024.

of day-to-day work decisions will be made autonomously through agentic AI by 2028.

Historical Foundation and Modern Catalyst

The deep roots of agentic AI



The evolution of agentic AI represents a fascinating journey that spans over six decades of computer science innovation. This evolution was followed by the rise of machine learning, which shifted the paradigm from handcrafted rules to statistical pattern recognition. These systems could learn from data and enabled advances in perception, classification, and decision-making.

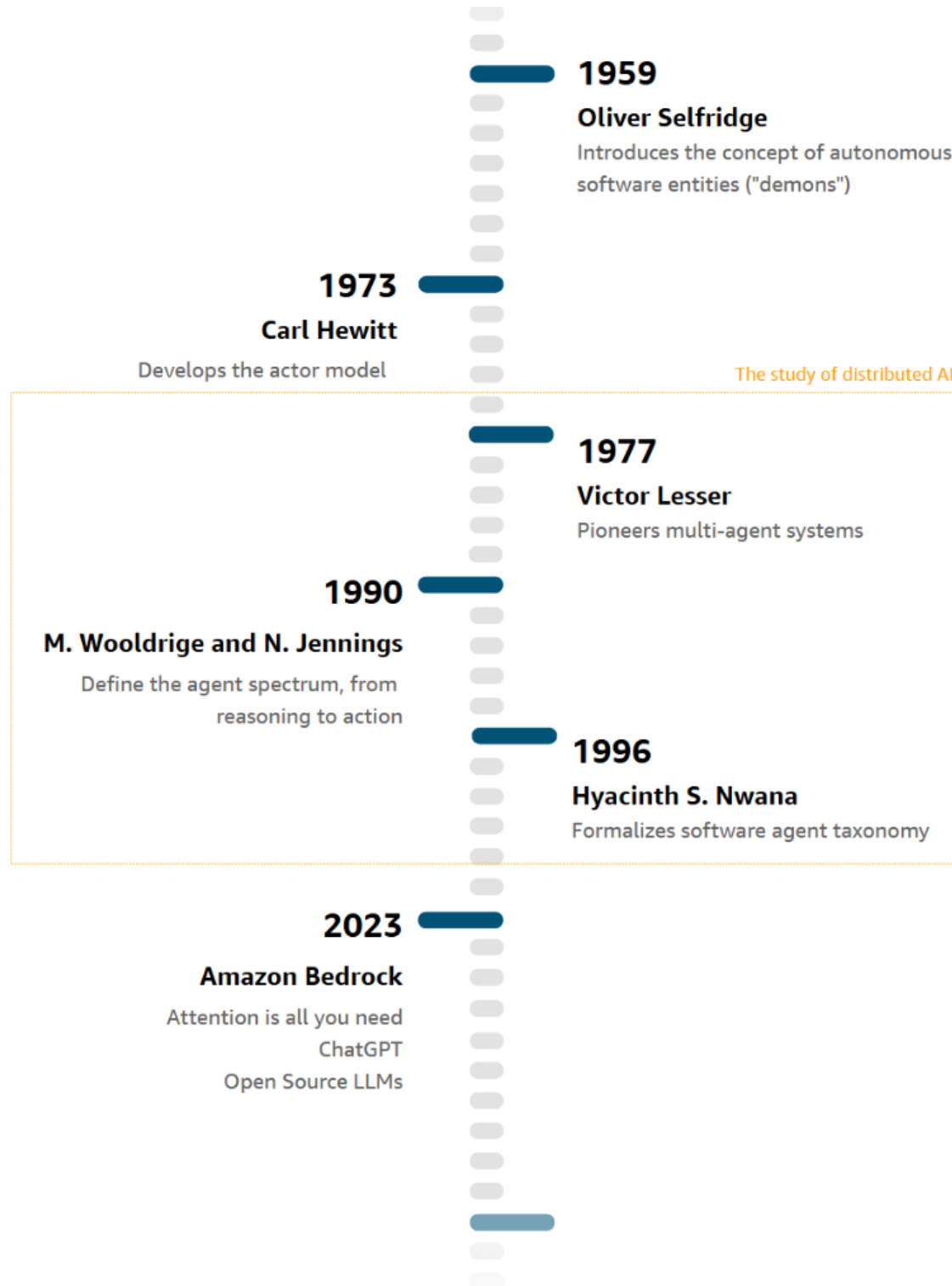
Starting in the late 1950s with Oliver Selfridge's groundbreaking concept of autonomous software entities, the field progressed through several crucial developments. The 1970s saw Carl Hewitt's actor model and Victor Lesser's multi-agent systems lay the foundational architecture for distributed intelligence, while the 1990s brought more sophisticated frameworks from Wooldridge, Jennings, and Nwana, who helped formalize how software agents should operate and interact.

Throughout the 2000s and 2010s, significant advances in machine learning, particularly deep learning and neural networks, laid crucial groundwork for agentic AI. The development of increasingly sophisticated language models and reinforcement learning techniques created new possibilities for agent autonomy and decision-making capabilities.

The real transformation occurred in the 2020s when these theoretical foundations merged with powerful new technologies. The introduction of enterprise-grade platforms like Amazon Bedrock Agents and the standardization through the Model Context Protocol (MCP) marked a significant leap forward. That platform is now Amazon Bedrock Agents Classic, closed to new customers since July 30, 2026. Today, Amazon Bedrock AgentCore is the recommended way to build and operate agents on AWS. By 2025, the field had matured considerably with the release of hybrid reasoning models and open-source frameworks like Strands Agents.

Today's agentic AI represents the convergence of multiple disciplines: the architectural principles of multi-agent systems, the cognitive capabilities of modern AI, the reasoning power of large language models, and the flexibility of cloud computing. This synthesis has created autonomous systems that can truly understand, reason, and act on behalf of users while continuously adapting and improving their performance.

Large language models (LLMs) represent a convergence of scale, architecture, and unsupervised learning. LLMs can reason, generate, and adapt to tasks with little or no task-specific training. By combining LLMs with scalable cloud-native infrastructure and composable architectures, we are now achieving the full vision of agentic AI: intelligent software agents that can operate with autonomy, context-awareness, and adaptability at enterprise scale.



The generative AI breakthrough



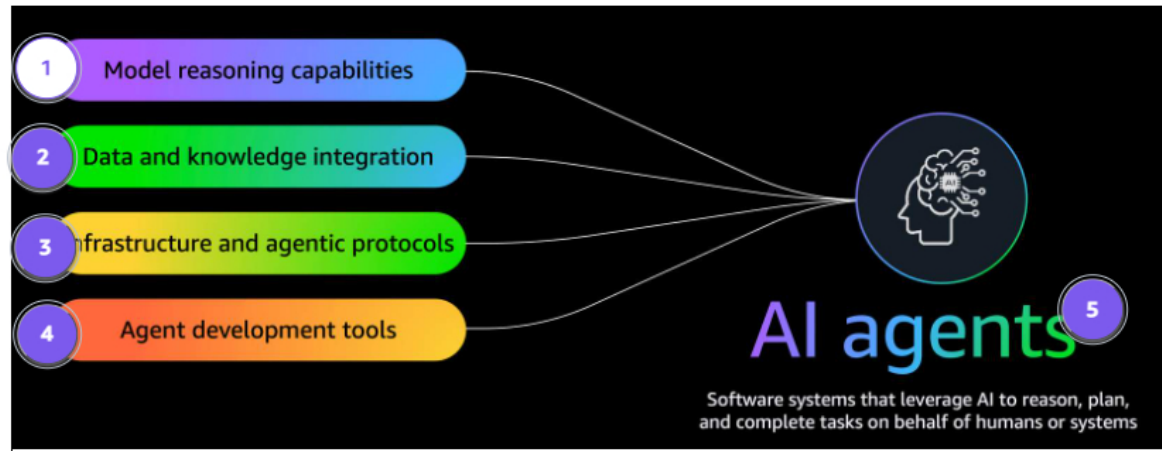
While software agents have evolved since the 1950s, **it was the breakthrough of generative AI** that finally made agentic AI practical for enterprise use. This combination brings together modern reasoning capabilities with proven distributed systems architecture, creating enterprise-ready solutions that are both innovative and reliable.

This evolution has produced intelligent systems, agentic AI, that can operate autonomously across the enterprise, handling everything from daily operations to complex business challenges. And what sets agents apart is their iterative thinking ability: they evaluate results, adjust their approach, and persistently work toward goals. Rather than simply answering questions, agents can solve complex problems through a process of exploration and refinement.

To transform agentic AI from hype to reality, AWS has identified four key enablers:

1. **Advanced model reasoning capabilities** provide smarter, more cost-effective foundation models.
2. **Robust data and knowledge integration** grounds agents in enterprise context and proprietary data.
3. **Scalable infrastructure and AI agent communication protocols** such as Model Context Protocol (MCP) and Agent-to-Agent (A2A) protocol to ensure secure agent operation and connectivity.
4. **Sophisticated agent development and orchestration tools** enable the design, deployment, and management of agents at scale.

These enablers collectively drive the practical implementation of agentic AI across industries and functions. Click on each marker to explore agentic AI enablers:



1: Model reasoning capabilities

Foundation models with advanced reasoning capabilities enable AI agents to understand context, make decisions, and solve complex problems.

AWS offers a range of cost-effective models optimized for different use cases, making sophisticated AI accessible for various enterprise needs.

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The Partner opportunity in agentic AI



For AWS Partners, this presents a compelling opportunity. Customers who have already embraced generative AI have a natural pathway to understanding and adopting agentic capabilities. Partners can leverage their customers' existing familiarity with generative AI to introduce more sophisticated agentic solutions. The implementation is enterprise-practical because it builds on proven foundations - distributed systems, multi-agent architectures, and cloud-native infrastructure - while incorporating the latest advances in AI reasoning and adaptation.

AWS provides Partners with both the innovation and the fundamentals needed to deliver successful agentic AI solutions. From Amazon Bedrock AgentCore to the Strands Agents framework, Partners have access to a comprehensive toolkit that combines cutting-edge AI capabilities with battle-tested distributed systems expertise. This unique combination enables Partners to guide their customers confidently from early experimentation to full-scale enterprise deployment of agentic AI solutions.

Beyond Generation to Action

The evolution from generative AI to agentic AI



To better illustrate the evolution from generative AI to agentic AI, let's compare two scenarios.

▼ **Scenario 1: Generative AI.** In this scenario, which you're likely familiar with, the AI assists by analyzing and generating content, but requires human direction for next steps.

"Analyze this quarterly sales report and summarize the key trends." - Generative AI prompt example

▼ **Scenario 2: Agentic AI.** In this case, the AI agent continuously monitors, makes decisions, and takes action autonomously to achieve a business goal.

"Monitor inventory levels, predict stock-outs, and maintain three months of inventory for critical products." - Agentic AI prompt example

The key difference? We're moving from AI that assists with specific tasks:

Analyze → Summarize

to AI that autonomously manages entire processes:

Analyze → Recommend → Plan → Execute → Monitor → Adapt

While generative AI excels at creating content and providing insights, agentic AI takes this further by understanding objectives, planning actions, and executing decisions over time. This represents a fundamental shift from assistance to autonomy - from tools that help us think to agents that help us do.

As Partners, understanding this evolution is crucial for helping customers move beyond simple AI assistance to implementing truly transformative autonomous solutions.

Understanding the autonomy spectrum

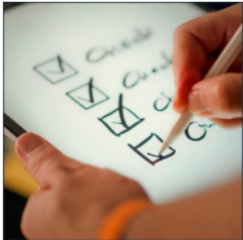


Let's explore how AI capabilities progress from simple, human-prompted tasks to autonomous, goal-driven behaviors. The key question is: how independently can the AI act? The following table shows how AI's scope expands as its autonomy increases, from performing single tasks to achieving full autonomy.

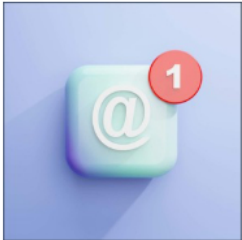
- ▶ Level 1: Single task, single response
- ▶ Level 2: Human-directed, connected actions
- ▶ Level 3: Contextual Understanding
- ▶ Level 4: Full Autonomy



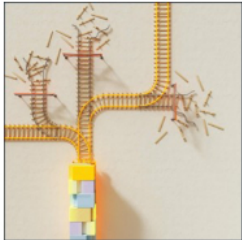
Responsive: AI performs a single task



Suggestive: AI is human-directed, provides suggestions



Proactive: AI automatically monitors



Autonomous: Agentic AI learns and adapts

Decision making evolution



Moving along the autonomy spectrum, we see a parallel evolution in decision-making capabilities. As AI becomes more autonomous, its ability to make and act on decisions also transforms.

Let's compare how generative AI and agentic AI approach decision-making across key characteristics:

	Generative AI	Agentic AI
Primary Role	Advises	Acts
Scope	Single-point analysis	End-to-end process
Decisions	Suggestions based on prompts	Autonomous choices within parameters
Timeline	Moment-in-time	Continuous monitoring and adjustment
Human Input	Required for each step	Sets goals and boundaries
Action Example	Suggests responses to customer queries	Handles entire customer interaction, escalating only when needed

As you help customers modernize their operations with AI, understanding these four key transitions in AI decision-making is crucial. Click through each stage to see how AI evolves from simple reactions to autonomous actions:

- ▶ Stage 1: From Reactive to Proactive Approach
- ▶ Stage 2: From Analysis to Execution Capabilities
- ▶ Stage 3: From Single Decisions to Decision Chains
- ▶ Stage 4: From Validation to Oversight

Context persistence



Ongoing learning and adaptation

While generative AI excels at individual interactions, agentic AI builds and maintains context over time, learning from each interaction to improve future outcomes.

💡 Agents need to maintain context across interactions.

Context persistence enables agents to make better decisions based on historical patterns, learn continuously from experiences, and maintain meaningful relationships with users. Without it, agents would be like a colleague who forgets everything overnight - functional but inefficient.

Let's explore this concept with an example.

Imagine a customer who writes: "I'm having issues with my cloud deployment."

▼ Scenario 1: Generative AI output. In this scenario, the interaction starts from scratch as the context must be manually provided:

"I'd be happy to help you with your cloud deployment issues.

To better assist you, could you please provide more details about the specific problem you're experiencing"

- Generative AI output example

▼ Scenario 2: Agentic AI output. In this scenario, the AI maintains ongoing context, building knowledge base over time:

"I see you're running a three-tier architecture on EC2. Last week we optimized your auto-scaling settings. Is the issue related to the recent changes?"

- Agentic AI output example



Navigating the AI agent conversation

As an AWS Partner, you're meeting with a retail company's CTO who successfully uses generative AI for customer service and content creation. During your review, they ask questions about AI agents.

 Customer question: *"What's the difference between our current AI assistant and these new 'agents' everyone talks about? We've invested significantly in AI already..."* **(Pick your response below to see what the customer says next.)**

 **Your response:** *"Unlike basic AI handling single tasks, agents use advanced frameworks to maintain memory and enable autonomous actions."*

 **Your response:** *"Your current AI assists with tasks on request. Agents proactively manage processes - like maintaining optimal inventory without supervision."*

 **Your response:** *"Your current AI is basically obsolete. Agents are the only way forward if you want to stay competitive in retail."*

 Customer follow-up question: *"Interesting. So, how do we control what it can do? How does it work?"*

(Pick your response below to see what the customer says next.)

 **Your response:** *"You set rules and limits. Agents operate within these, like a manager following policy. They learn and adapt, but always within your guidelines."*

 **Your response:** *"Control is implemented via hierarchical reinforcement learning algorithms and multi-agent coordination protocols within defined action spaces."*

 **Your response:** *"Agents are fully autonomous. Once set up, they handle everything without any need for human oversight or intervention."*

Excellent. Next, for example, the customer could be interested in exploring a pilot project to demonstrate how agents can optimize a specific area of their retail operations.

Activity: Rehearse your agentic AI explanation 🎯

🕒 Time: 2-5 minutes

👉 **Format:** Individual reflection or practice with a peer

🎯 **Outcome:** Confident, business-focused explanation of agentic AI value

📄 **Instructions:**

- Review the content in this course so far
- Create two business-focused explanations of agentic AI: one for customers already familiar with AI, and one for those new to generative AI. In both versions, emphasize business value over technical details and include a relevant industry example.

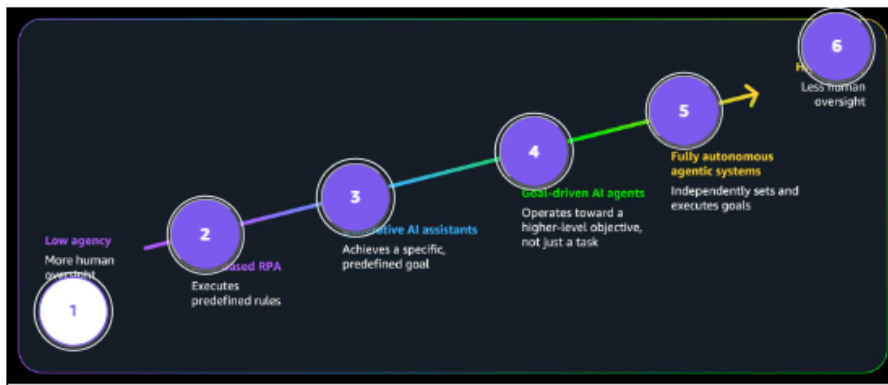
The Maturity Progression

Introduction

As organizations adopt agentic AI, they typically progress through distinct maturity levels. Understanding this progression helps avoid 'agent washing' - the practice of mislabeling basic automation as 'AI agents' without true autonomous capabilities - and ensures sustainable implementation.

The Agentic AI Maturity Scale illustrates the enterprise journey from simple task automation to fully autonomous systems. The scale progresses from low-agency systems requiring significant human oversight on the left, to high-agency systems capable of independent operation and strategic decision-making on the right.

Click on each marker to explore the Agentic AI Maturity Scale:



1: Low Agency

- Requires constant human oversight
- Follows fixed, predefined rules
- Handles single, specific tasks

< >

Most organizations begin their journey with rule-based RPA, automating simple, repetitive tasks, then progress to generative AI assistants that can handle broader workflows. While some are experimenting with goal-driven agents, fully autonomous systems remain a future aspiration for most enterprises. Reaching higher agency requires not just better technology but also governance, trust, and organizational readiness.

Activity: map your customer agentic AI maturity

Time: 10-15 minutes

Format: Individual reflection

Outcome: Getting a clear understanding of your customers' current maturity levels and next steps

Instructions:

- Think of 3-5 customers you're currently working with;
- For each customer, consider their current AI implementations (the level of human oversight required, the ability to handle complex tasks, decision-making autonomy);

Knowledge check



1. Which of the following is AWS's definition of AI agents?

AI agents are chatbot interfaces that use natural language processing to answer customer



AI agents are software programs that use machine learning algorithms to analyze data and generate



AI agents are autonomous software systems that leverage AI to reason, plan, and complete tasks on behalf



AI agents are digital assistants that coordinate between different AI models to process user requests and



2. A customer service AI system remembers previous interactions with a customer and uses this information to provide more personalized responses. This demonstrates:

✗ Incorrect.

✗ Incorrect.

✓ Correct.
Context persistence enables AI agents to build meaningful relationships by remembering



✗ Incorrect.

Multi-Agent Systems and Agent Types

Multi-agent systems

Amazon Bedrock multi-agent collaboration allows developers to build, deploy, and manage multiple specialized agents seamlessly working together to address increasingly complex business workflows. Each agent focuses on specific tasks under the coordination of a supervisor agent, which breaks down intricate processes into manageable steps to ensure precision and reliability.

Multi-agent systems excel at valuable tasks that involve heavy parallelization, information that exceeds single context windows, and interfacing with numerous complex tools.

Agent types

As agentic AI evolves, different types of agents emerge with varying levels of sophistication and autonomy. Understanding these types and how they work together is crucial for building effective solutions.

Reactive Agents

Deliberate Agents

Goal-driven Agents

- Reactive agents **respond to specific triggers with predefined actions** ("If This... Then That")
- No complex reasoning
- Example: "When inventory drops below 100 units, order 500 more"
- AWS Context: **Simple agents on Amazon Bedrock AgentCore Runtime with rule-based triggers**

Reactive agents are like thermostats, instantly responding to temperature changes.



Multi-agent coordination



Now that we understand the different types of agents, let's explore how they work together. Just as a business combines different roles and skills, multiple agents can coordinate in various patterns to achieve complex goals.

For example, in an e-commerce system built with AWS, a reactive agent running on Amazon Bedrock AgentCore Runtime monitors inventory levels, reaching Amazon S3 and DynamoDB as tools through AgentCore Gateway; a deliberate agent leverages foundation models in Amazon Bedrock to analyze pricing and demand patterns; and a goal-driven agent manages the overall customer experience, using AgentCore Memory to keep context across sessions.

Agents can work together using three main coordination patterns:

Agent handoffs

Like in a relay race: the inventory agent passes to the payment agent, then to the shipping agent.

< Here we have a sequential task completion: >

Agent A → Agent B → Agent C

< 1 2 3 >

Multi-agent systems combine specialized expertise, complex problem-solving, and collaborative accuracy to deliver better results than single agents working alone.

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Parallel processing

Like a restaurant kitchen, here multiple agents handle different tasks simultaneously.

< Agent A →

Agent B →

Agent C →



< 1 2 3 >

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Hierarchical systems

In a management structure, a goal-driven agent oversees and coordinates specialized agents.



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Agent Reliability and Limitations

Understanding agent challenges



While AI agents offer powerful capabilities, understanding their limitations and implementing proper safeguards is important for successful deployment.

Let's explore common challenges and their solutions.

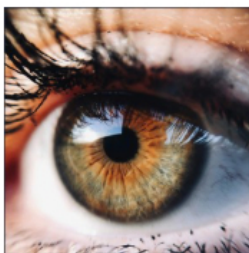
Agent hallucinations and errors



Agent hallucinations occur when AI agents 'make things up' or take actions that don't match reality - similar to a GPS confidently directing you to a non-existent street or a translation app creating nonsensical phrases. Agent hallucinations happen when AI generates responses that seem plausible but are factually incorrect or when it takes actions based on misinterpreted information, potentially impacting business operations.

Hallucinations have a potentially high impact on end users, as the system might provide wrong product information or incorrect pricing. Therefore, we need systems to verify agent actions and reduce hallucination risks as much as possible.

To mitigate these risks, several tools are available: human oversight, validation checkpoints, and confidence thresholds.



Human Oversight: Having people review and approve critical agent decisions before they're executed - like a manager reviewing price changes before they go live.



Validation Checkpoints: Building automatic verification steps that compare agent responses against trusted data sources - similar to a spell-checker flagging potential errors.



Confidence Thresholds: Setting minimum confidence levels that agents must meet before taking action - like requiring 90% certainty before processing a high-value transaction.

AWS implements these safeguards through [Amazon Bedrock Guardrails](#) and [AgentCore](#). With Amazon Bedrock AgentCore's monitoring capabilities, you can ensure the AI agents perform reliably and efficiently by tracking, troubleshooting, and optimizing their operations in real-time.

[Amazon Bedrock Guardrails](#) provides configurable safeguards to help safely build generative AI applications at scale. With a consistent and standard approach used across a wide range of foundation models (FMs) including FMs supported in Amazon Bedrock, fine-tuned models, and models hosted outside of Amazon Bedrock, Guardrails delivers industry-leading safety protections:

[AgentCore Observability](#), for example, provides operational visibility into AI agent for monitoring, debugging, and supporting compliance. Integrates with preferred dashboards via OpenTelemetry compatibility.

[AgentCore Evaluations](#) closes the loop on reliability. It is a purpose-built evaluation service that measures how well your agents and tools complete tasks, handle edge cases, and keep their output reliable across different inputs and contexts. You can run evaluations before you deploy, to catch weak reasoning and hallucination risk early, and continuously after deployment, so quality is measured rather than assumed. Results flow into AgentCore Observability, giving you one place to see both what an agent did and how well it did it.

Coordination failures



When multiple AI agents work together, they can accidentally interfere with each other's tasks, like teammates working without coordination. Imagine two customer service agents trying to help the same customer at once: Agent A updates the customer's email address. Agent B simultaneously updates the same customer's phone number. Without proper coordination, one update might overwrite the other, or create conflicting records.

Coordination failures can lead to duplicate actions, conflicting updates and results, and wasted resources.

To prevent agents from working at cross-purposes, there are three key solutions: Consensus protocols, conflict resolution, state management.



Consensus Protocols: Think of it like a team meeting protocol: before making changes, agents must 'check in' with each other. Like coworkers using a shared calendar to avoid scheduling conflicts.



Conflict Resolution: It works like a traffic light system: it determines which agent has priority, sets rules for who goes first, manages any conflicting changes automatically.



State Management: It acts like a central record-keeper, as it maintains the 'single source of truth'. Like a bank tracking your account balance across multiple transactions, it ensures all agents work with the latest information.

AWS implements these through AgentCore's coordination services, helping agents work together smoothly.

How to avoid costly mistakes?



Customers frequently ask how to prevent agents from making costly mistakes. The key question is how to effectively contain and mitigate potential errors in agent operations. There are different ways to address this: setting boundaries and enabling quick rollback capabilities if things don't go as expected. Here are some techniques to ensure **operational stability**:

▼ Setting Confidence Scoring

Agents must be highly confident before making decisions.

For example: "If an agent is less than 90% confident about a pricing decision, it must seek human approval"

▼ Establishing Approval Thresholds

Set financial limits for autonomous decisions.

For example: "Agents can process refunds up to \$100 automatically, but need approval for larger amounts"

▼ Configuring Rollback Capabilities

Integrate the ability to quickly undo agent actions if needed.

For example: "If an incorrect price is posted, the system can automatically revert to the previous verified price"

By implementing these guardrails, organizations can protect against costly mistakes while maintaining agent productivity through controlled automation and clear safety measures.

How to deploy reliable agents?



AWS recommends to deploy agents gradually: start simple, maintain oversight, monitor closely, and increase autonomy as reliability improves.

✓ Dos

Start Simple: Begin with basic reactive agents that handle straightforward tasks; gradually introduce more complex functionalities.

Human Oversight: Implement approval workflows for high-impact decisions; ensure human review of critical actions.

Monitoring: Set up continuous performance tracking; implement robust error detection mechanisms; regularly analyze agent behavior and outcomes.

Gradual Autonomy: Incrementally increase agent independence; base autonomy increases on demonstrated reliability; expand agent capabilities as trust in their performance grows.

By following these practices, organizations can deploy AI agents responsibly, minimizing risks while maximizing the benefits of automation and AI-driven decision-making.

Scale Requires Operational Excellence

Introduction

Proofs of Concept (PoCs) serve as foundations for successfully scaling agentic AI systems and achieving operational excellence. They provide organizations with invaluable insights by creating controlled testing environments where potential failure points, scalability bottlenecks, and performance issues can be identified early in the development cycle.

The systematic approach of PoCs, which typically progresses through planning, implementation, evaluation, and recommendation phases, ensures that organizations can scale their agentic AI implementations while maintaining operational excellence and minimizing risks. This methodical validation process encompasses technical aspects (such as agent coordination and resource utilization), operational considerations (including monitoring requirements and maintenance procedures), and business validation (covering value proposition and cost structure). By starting small, measuring comprehensively, testing edge cases, and documenting learnings, organizations can leverage PoCs to create robust, scalable, and operationally excellent agentic AI systems that deliver sustainable business value.



Discover the journey from AI agent proof-of-concept to enterprise-scale deployment, ensuring operational success and business value. (Duration 01:38)

Transcript

development and deployment, while scalability features ensure consistent performance under varying loads.

These fundamental elements transform an initial POC into a reliable, scalable business solution.

Remember, the gap between POC and production requires addressing all these challenges systematically to

build trustworthy, scalable systems that deliver consistent business value."

AWS Prescriptive Guidance (pages 3-32)

Guide your customers through their generative AI adoption journey through a structured four-level framework to assess, guide, and transform clients' AI capabilities across business, technical, and operational dimensions.

[Read PDF.](#)

Knowledge check



According to the Agent Classification Framework, which of the following correctly represents the progression of agent types, from simplest to most complex?

✗ Incorrect.

✗ Incorrect.

✗ Incorrect.

✓ Correct. The AWS taxonomy of AI agents consists of reactive agent which respond to specific triggers.



The Three Value Categories

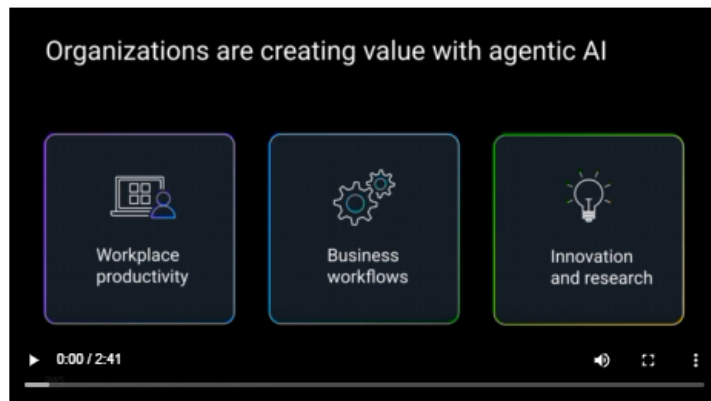
Introduction

As enterprises increasingly recognize the transformative potential of generative AI, they are strategically investing in AI agents across three distinct but interconnected value categories that promise to revolutionize how organizations operate, innovate, and compete in the digital age.

- ▶ Workplace productivity agents
- ▶ Business workflow agents
- ▶ Innovation and research agents

Together, these three categories of AI agents represent an approach to enterprise AI adoption addressing immediate operational needs while simultaneously driving long-term strategic advantages. Organizations are doubling down on these agents not just for incremental improvements but for the transformative potential they offer in reimagining how work gets done, how processes flow, and how innovation occurs in the modern enterprise.

The integration of these agents with other enterprise systems and databases has further enhanced their capabilities. They can now seamlessly interact with Customer Relationship Management (CRM) systems, Enterprise Resource Planning (ERP) platforms, and various productivity tools, allowing for more comprehensive and efficient task completion.



See how customers across industries use agentic AI to transform productivity, automate workflows, and accelerate innovation through real-world examples. (Duration 2:41)

Let's dive deeper in each category and case study.



Transcript

AWS has been working with customers across virtually every industry on their generative AI and agentic

AI strategies, and we're seeing agentic AI creating business value in three key areas:

Workplace productivity: Representing the majority of current agentic AI applications, these agents seamlessly

integrate with existing tools like Zoom or Asana, or a developer's IDE automating routine tasks and

Workplace productivity agents



In the early stages, workplace productivity agents were primarily designed to automate routine, repetitive tasks such as scheduling meetings, managing emails, or basic data entry. These systems relied heavily on predefined rules and workflows, offering limited flexibility and requiring significant human oversight.

As natural language processing (NLP) and machine learning technologies advanced, we saw the emergence of more intelligent virtual assistants. These agents could understand and respond to natural language queries, provide more contextual support, and learn from user interactions to improve their performance over time. However, they still largely depended on human input and guidance for most tasks.

The introduction of large language models (LLMs) and transformer architectures marked a significant leap forward. Agents powered by these technologies demonstrated ability to understand context, generate human-like text, and tackle more complex cognitive tasks. This advancement enabled workplace productivity agents to move beyond simple query-response interactions to more sophisticated task completion capabilities like:

- Understand and break down complex tasks into manageable steps
- Access and synthesize information from multiple sources
- Make decisions based on predefined criteria and learned patterns
- Execute actions across various software applications and platforms
- Adapt to new situations and learn from past experiences

Workplace productivity multi-agent patterns



A multi-agent pattern is an architectural approach where multiple AI agents collaborate and interact with each other to accomplish complex tasks. It involves a system where different specialized AI agents work together, each handling specific aspects of a larger task or workflow. These agents communicate, coordinate, and sometimes even negotiate with each other to achieve common goals.

Below you can find some examples of agents that can enhance workplace productivity:

- ▶ Document processing agents
- ▶ Training coordination agents
- ▶ Progress tracking agents

Case study



NFL's game-changing generative AI transformation

The NFL leveraged AWS to drive innovative fan, player, and employee experiences powered by cutting-edge generative AI services like Amazon SageMaker AI and Amazon Bedrock.

[Read the full customer story.](#)

Key figures

- 67% reduction in new hire training time
- 50+ AI/ML Next Gen Stats built with AWS, including 10 new stats for the 2024 season
- 10,000 seasons' worth of plays analyzed to better understand potential risks to players
- 1M+ data points analyzed during every NFL game

Business workflow agents



The evolution of generative AI in business workflow agents represents a fundamental shift from basic process automation to intelligent, autonomous process orchestration. This transformation has redefined how organizations approach their operational workflows, moving from rigid, rule-based systems to adaptive, intelligent solutions capable of understanding, managing, and optimizing complex business processes.

Initially, workflow automation relied on predefined paths and conditional logic, requiring extensive human configuration and intervention. Early systems could follow predetermined steps but lacked the intelligence to handle exceptions or adapt to changing conditions. As previously mentioned, the introduction of generative AI marked a pivotal transition, enabling workflow agents to understand process context, learn from historical data, and make intelligent decisions autonomously.

Modern business workflow agents demonstrate sophisticated capabilities in both process assistance and autonomous execution:

- ▶ **Process assistance evolution**
- ▶ **Autonomous process execution advancement**

The latest developments in this field show workflow agents evolving into sophisticated process partners that can:

- Understand natural language instructions and translate them into executable workflows
- Automatically adjust processes based on changing business conditions
- Coordinate multiple sub-processes across different departments and systems
- Ensure compliance while maintaining process efficiency
- Generate insights for continuous process improvement

This evolution has enabled organizations to achieve high levels of operational efficiency, reducing manual intervention while improving accuracy and consistency in business processes. The future trajectory suggests even greater autonomy, with business workflow agents becoming increasingly capable of managing complex, cross-functional processes while maintaining the flexibility to adapt to changing business needs.

Business workflow multi-agent patterns



Here are examples of agents that can enhance business workflows:

- ▶ **Data collection agents**
- ▶ **Analysis agents**
- ▶ **Compliance report agents**
- ▶ **Cross-functional agents**

Case study



Multi-agent AI: Realize the next frontier of enterprise performance

Transformed business processes by rapidly building and scaling agentic AI systems.

[Read the full customer story.](#)

Innovation and research agents



Innovation and research agents represents a paradigm shift in how organizations approach discovery, experimentation, and knowledge creation. From basic research assistance tools to autonomous systems capable of hypothesis generation and testing, these agents have transformed the innovation landscape.

Initially, research agents served primarily as information retrieval and organization tools, helping researchers gather and categorize existing knowledge. The introduction of generative AI marked a revolutionary change, enabling agents to not only process existing information but also generate new insights, hypotheses, and potential solutions. This transition has led to the emergence of two distinct but interconnected capabilities:

- ▶ **Research assistance evolution**
- ▶ **Autonomous discovery and testing advancement**

Modern innovation and research agents demonstrate capabilities in:

- Generating novel research directions by identifying patterns in unrelated fields
- Conducting virtual experiments to test hypotheses rapidly
- Analyzing and interpreting complex research data
- Proposing and validating new theoretical frameworks
- Accelerating the research cycle through automated testing and validation

These advancements have reduced the time and resources required for innovation while expanding the scope of what's possible in research and development. The integration of machine learning, natural language processing, and specialized domain knowledge has created agents that can work alongside human researchers, augmenting their capabilities and accelerating the pace of discovery.

Innovation and research multi-agent pattern



Below you can find some examples of agents that can enhance innovation and research:

- ▶ **Data gathering agents**
- ▶ **Analysis agents**
- ▶ **Recommendation agents**
- ▶ **Implementation agents**

Customer Archetype Recognition

What is customer archetype recognition?



Customer archetype recognition is a specialized AI capability where agents autonomously identify, learn, and adapt to different customer patterns, behaviors, and interaction styles in real-time, enabling dynamic and personalized engagement through multi-agent systems.

This agentic approach to customer archetype recognition enables:

- More sophisticated customer understanding
- Dynamic interaction adaptation
- Improved customer experience
- Enhanced personalization at scale
- Better resource utilization
- Increased engagement effectiveness

This capability enables organizations to move beyond traditional demographic segmentation to more sophisticated, behavior-based customer understanding and engagement.

Customer journey patterns:

- 1 -



Agent-ready software and APIs

Adapt existing software to be used autonomously by AI agents

How do we make sure AI agents can easily discover and use our software tools, just like humans do?

The goal is making your tools as self-describing as possible - AI agents should understand what they do, how to use them, and what to expect, just from the interface itself.

- ▶ API-first design
- ▶ Structured schemas
- ▶ Function calling standards
- ▶ Discovery mechanisms
- ▶ Error handling

What do protocols like MCP (Model Context Protocol) and A2A (Agent-to-Agent) communication mean for our business?

These protocols are already changing how businesses operate and compete. Both are established industry standards today, and Amazon Bedrock AgentCore supports MCP, A2A, and AG-UI natively.

▼ MCP (Model Context Protocol)

- **Standardized AI integration:** Your tools become instantly compatible with any MCP-enabled AI system
- **Reduced integration costs:** No custom connectors for each AI platform
- **Ecosystem participation:** Join a broader network of interoperable tools
- **Competitive advantage:** Being MCP-ready is now a baseline expectation, and it gains you distribution through AI marketplaces

▼ A2A (Agent-to-Agent Communication)

- **Automated partnerships:** AI agents negotiate and execute business processes autonomously
- **Dynamic supply chains:** Real-time optimization without human intervention
- **New revenue models:** Usage-based pricing, micro-transactions between agents
- **Operational efficiency:** 24/7 automated business operations

- 2 -



Embedded agentic SaaS and APIs

Streamline user/business workflows and processes inside of existing SaaS/apps (transparent to end users)

How can we automate repetitive tasks behind the scenes without our users even knowing AI is involved?

The best AI automation is invisible - users just experience better, faster results without knowing how it happens.

Invisible Automation Strategies:

▼ Background processing

- Auto-optimize content while users work on other things
- Pre-generate likely next steps based on user patterns
- Continuously improve recommendations without user input

▼ Predictive actions

- Cache content before users request it
- Auto-correct common mistakes silently
- Suggest improvements without interrupting workflow

▼ Seamless enhancement

- Improve search results quality invisibly
- Auto-tag and organize user content
- Generate summaries and insights automatically

▼ Implementation principles

- Make the primary action instant (pre-computed results)
- Handle edge cases gracefully without exposing complexity
- Provide escape hatches for power users who want control
- Log improvements for transparency if questioned

Users should think the software is just really well-designed, not that AI is doing the heavy lifting. The magic happens in the margins - faster responses, better suggestions, fewer errors - without any AI branding or explanations needed.

How can we infuse AI-driven decision making abilities into our existing static workflows?

Transform static workflows into adaptive systems by adding decision points where AI evaluates context and chooses the optimal path forward.

Integration strategies:

▼ Smart routing

- Add AI decision gates at workflow branches
- Route requests to different processing paths
- Maintain existing logic as fallback options

▼ Dynamic parameters

- AI adjusts parameters based on real-time context
- Optimize timeouts, retry logic, and resource allocation
- Personalize outputs without changing core workflow

▼ Adaptive orchestration

- AI reorders workflow steps based on efficiency
- Skip unnecessary steps when conditions are met
- Parallel processing decisions based on resource availability

▼ Context-aware branching

- Inject user history, system load, and business rules into decisions
- A/B test different AI decision strategies
- Gradually increase AI control as confidence grows

▼ Risk management

- Always maintain manual override capabilities
- Log AI decisions for audit and improvement
- Gradual rollout with performance monitoring
- Clear rollback procedures for each enhancement

Start with low-risk decisions and gradually expand AI control as you build confidence in the outcomes.

How can we handle intelligent workflows automatically without having to hire more people?

Automate the cognitive work, not just the repetitive tasks - AI can handle complex decision-making that previously required human judgment.

Intelligent automation strategies:

▼ Autonomous decision making

- **Quality Control:** AI evaluates output quality and decides on acceptance/rejection
- **Resource Allocation:** AI distributes workload based on system capacity and priorities
- **Exception Handling:** AI diagnoses issues and applies appropriate fixes
- **Performance Optimization:** AI continuously tunes workflow parameters

▼ Workflow orchestration

- AI decides optimal task sequencing
- Automatic parallelization when beneficial
- Smart queuing and priority management

▼ Customer support

- AI handles routine inquiries end-to-end
- Escalates complex issues with full context
- Learns from resolution patterns

▼ Operations management

- Predictive maintenance scheduling
- Automatic capacity planning
- Intelligent alerting (reduces noise)

How can we increase customer value (either via acquisition or retention) by educating our customers on how to work with AI agents directly, using the language of their choice?

Teaching customers to work effectively with AI agents creates competitive moats and increases lifetime value through deeper platform engagement.

Value creation strategies:

▼ **AI Literacy as product differentiation**

- Offer AI Prompt Engineering courses
- Create certification programs for AI agent collaboration
- Position your company as the AI-native solution provider

▼ **Multi-lingual AI agent interface**

- Develop AI agents capable of understanding multiple languages
- Implement real-time translation capabilities
- Ensure cultural nuances are respected in interactions

▼ **Personalized onboarding experience**

- Create interactive tutorials tailored to each customer's language
- Develop step-by-step guides for AI agent interaction
- Implement gamification elements to encourage learning

▼ **Natural language command library**

- Provide a comprehensive list of commands in multiple languages
- Create context-sensitive suggestions for commands
- Develop a searchable database of common queries and responses

▼ **AI interaction sandbox**

- Create safe environments for customers to practice AI interactions
- Provide simulated scenarios for different use cases
- Offer feedback and improvement suggestions

▼ **Continuous learning prompts**

- Implement AI-driven suggestions for new features or commands
- Provide context-aware tips during customer interactions
- Develop a "skill of the day" feature in the customer's language

▼ **Integration workshops**

- Host virtual or in-person workshops in multiple languages
- Demonstrate how to integrate AI agents into existing workflows
- Provide hands-on experience with AI tools

The customers who learn to leverage AI through your platform become your most valuable, sticky users - they've invested in skills that are tied to your ecosystem.

How should we price our products when AI agents are using them instead of human users or traditional API calls?

AI agents fundamentally change pricing dynamics - they operate at machine speed, have different value perceptions, and enable new business models.

Key pricing considerations:

▼ **Volume vs value shift**

- AI agents generate 10-100x more requests than humans
- But each individual request may have lower perceived value
- Focus on outcome-based pricing rather than per-call pricing

▼ **Agent-specific tiers**

- **Exploration tier:** Low cost for AI agents testing capabilities
- **Production tier:** Higher cost but guaranteed SLAs and priority
- **Enterprise tier:** Bulk pricing for high-volume agent deployments

▼ **Value-based models**

- Price based on business outcomes (courses completed, students trained)
- Revenue sharing for agents that generate end-user value
- Success fees for measurable improvements

▼ **Dynamic pricing strategies**

- Real-time pricing based on demand and compute costs
- Auction-based pricing for premium resources during peak times
- Loyalty discounts for consistent, predictable usage patterns

▼ **Risk mitigation**

- Rate limiting to prevent runaway costs
- Circuit breakers for unusual usage patterns
- Clear cost estimation tools for agent developers

The goal is making your service economically attractive for AI agents while capturing fair value from the outcomes they enable.

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Agent-facing agents

Orchestrate multi-agent systems to achieve complex tasks

Getting started:

▼ Phase 1: Single agent foundation

- Build one specialized agent
- Implement standard communication protocols (MCP/Open API format)
- Add self-description and capability discovery
- Test with simple single-step tasks

▼ Phase 2: Agent coordination

- Create orchestrator that can chain multiple agents
- Implement task decomposition and delegation
- Add error handling and retry logic
- Build context passing between agents

▼ Phase 3: Intelligent workflows

- AI-powered workflow planning
- Dynamic agent selection based on capabilities
- Parallel execution and dependency management
- Learning from workflow outcomes

Start simple with one well-designed agent, then expand the ecosystem as patterns emerge.

How do we make sure other AI systems can easily discover, use, monitor, and understand our AI Agents?

Create comprehensive agent metadata and standardized interfaces that make your agents self-documenting and observable. Focus on interoperability, standardization, and transparency.

Integration Standards:

- **Open API specs** for REST endpoints
- **MCP protocol** for local tool integration

▼ Standardized agent descriptions

- Use machine-readable formats (JSON, YAML)
- Implement semantic web technologies
- Adopt industry-standard schemas

▼ Agent discovery mechanisms

Registry system:

- Centralized agent directory
- Decentralized discovery protocols
- Semantic search capabilities

Discovery protocols:

- Implement service discovery (e.g., mDNS, UPnP)
- Use well-known endpoints for self-description
- Support for query-based discovery

Metadata:

- Tags for easy categorization
- Semantic annotations for capabilities
- Performance metrics and usage statistics

▼ Standardized APIs and interfaces

API standards:

- RESTful API design
- GraphQL for flexible querying
- gRPC for high-performance communication

Documentation:

- Open API specifications
- Interactive API documentation
- Code samples in multiple languages

Versioning:

- Clear versioning strategy
- Backward compatibility guidelines
- Deprecation policies and timelines

▼ Monitoring and observability

Telemetry:

- Expose standardized metrics (e.g., Prometheus format)
- Implement distributed tracing (e.g., OpenTelemetry)
- Provide logging interfaces (e.g., structured logs)

Health checks:

- Implement standard health check endpoints
- Provide status and readiness probes
- Expose performance and resource usage metrics

Alerting:

- Define standard alert formats
- Implement callback mechanisms for notifications
- Support subscription to status changes

▼ Semantic understanding

Ontology:

- Develop or adopt standard ontologies for agent capabilities
- Use semantic web technologies for knowledge representation
- Implement reasoning capabilities for inferring agent properties

Context sharing:

- Standardized context representation
- Mechanisms for sharing and updating context
- Support for context-aware interactions

▼ Interoperability protocols

Communication:

- Support for standard AI agent protocols (e.g. ACP, MCP, A2A, AG-UI)
- Implement publish-subscribe mechanisms
- Enable event-driven architectures

Data exchange:

- Use standard data formats (e.g. JSON)
- Implement data validation (JSON Schema)
- Support for data transformation and mapping

Security:

- Standardized authentication methods (OAuth, JWT)
- Implement fine-grained authorization
- Support for encrypted communications (TLS)

Knowledge check



1. Which of the following best describes the evolution of workplace productivity agents?

☐ Incorrect.	☐ Incorrect.	☒ Correct. The text describes the evolution of workplace productivity agents from simple rule-based systems.	☐ Incorrect.
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2. What is a key characteristic of business workflow agents according to the lesson "The Three Value Categories"?

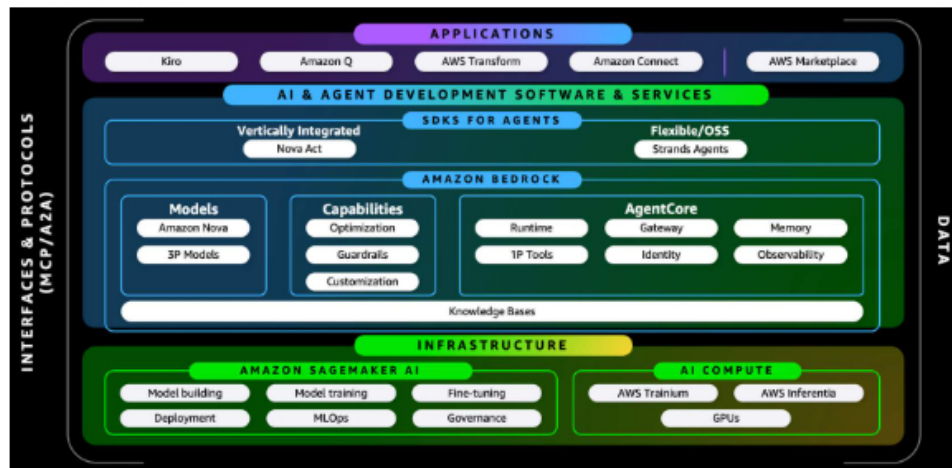
☐ Incorrect.	☐ Incorrect.	☒ Correct. The text explicitly states that "Business workflow agents take automation to the next level by orchestrating	☐ Incorrect.
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3. In the context of Customer Archetype Recognition, what does the text suggest about AI agents?

☐ Incorrect.	☐ Incorrect.	☐ Incorrect.	☒ Correct. The passage defines Customer Archetype Recognition as "A specialized AI capability where agents
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The Complete Architecture

AWS AI stack



Applications

- ▶ Kiro
- ▶ Amazon Quick
- ▶ AWS Transform
- ▶ Amazon Connect
- ▶ AWS Marketplace

SDKs for agents



[Amazon Nova Act](#) - an AI model trained to perform actions within a web browser.

Strands Agents



Strands Agents is an open-source SDK designed for building autonomous AI agents with a model-first approach. It provides a flexible, extensible framework designed to work seamlessly with AWS services while remaining open to integration with third-party components. Strands Agents is ideal for building fully autonomous solutions.

Key features of Strands Agents



Strands Agents includes the following key features:

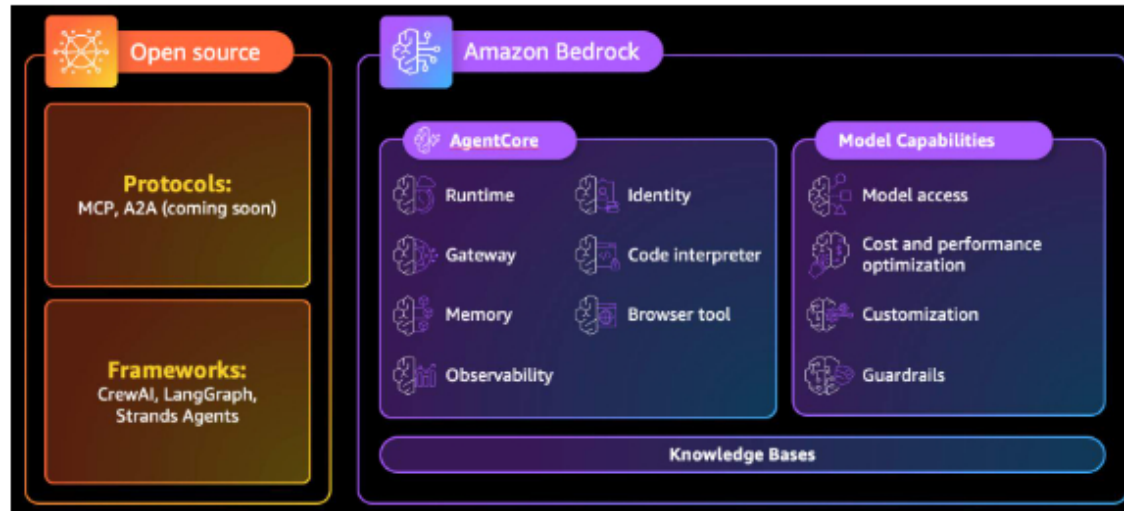
- ▶ Model-first design
- ▶ MCP integration
- ▶ AWS service integration
- ▶ Foundation model selection
- ▶ LLM API integration
- ▶ Multimodal capabilities
- ▶ Tool ecosystem

When to use Strands Agents



Strands Agents is particularly well-suited for autonomous agent scenarios including:

1. Organizations that build on AWS infrastructure who want native integration with AWS services for autonomous workflows
2. Teams that require enterprise-grade security, scalability, and compliance features for production autonomous systems
3. Projects that need flexibility in model selection across different providers for specialized autonomous tasks
4. Use cases that require tight integration with existing AWS workflows and resources for end-to-end autonomous processes

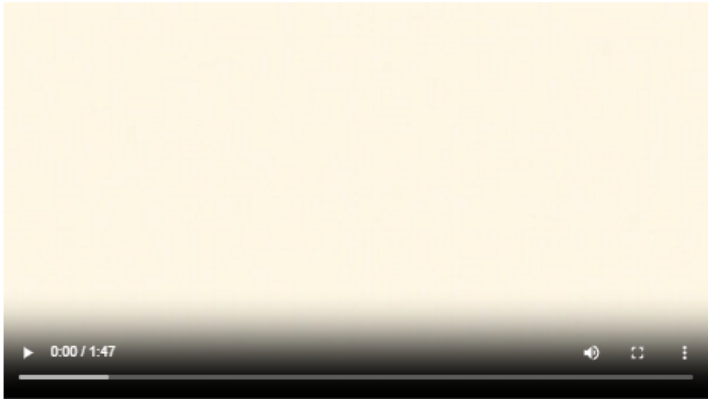


Amazon Bedrock is a comprehensive, secure, and flexible service for building generative AI applications and agents. Amazon Bedrock connects you to leading foundation models (FMs), services to deploy and operate agents, and tools for fine-tuning, safeguarding, and optimizing models along with knowledge bases to connect applications to your latest data so that you have everything you need to quickly move from experimentation to real-world deployment.

- **Model choice:** Amazon Bedrock gives access to over one hundred FMs from leading AI companies along with evaluation tools to pick the best model based on your unique performance and cost needs. Future-proof your AI strategy as your customer's needs evolve and new models emerge.
- **Customize with data:** Move from generic AI to AI that knows your customer and your business by customizing models with your data. By combining multiple data customization tools—Knowledge Bases, Bedrock Data Automation, prompt engineering, and fine-tuning—you can optimize your AI applications to your business, while ensuring you're always in control of sensitive information.
- **Safety and guardrails:** Bedrock Guardrails can help block up to 88% of harmful content and identify correct model responses with up to 99% accuracy to minimize hallucinations and data ambiguity using Automated Reasoning checks. Bedrock never stores or uses your data to train models, ensuring complete security and privacy, with encryption of data in transit and at rest, as well as identity-based policies for managing data access. Bedrock provides comprehensive monitoring and logging capabilities that can support your governance and audit requirements.
- **Cost optimization:** Ensure your AI applications are optimized for the perfect balance of cost, speed, and accuracy. Features like Model Distillation, Prompt caching, and Intelligent Prompt Routing can reduce expenses while maintaining performance.
- **Orchestration:** With Amazon Bedrock AgentCore you can build, deploy, and operate AI agents quickly and securely, using any framework and any model. Amazon Bedrock Flows provides a visual builder tool to connect Prompts, Guardrails, Knowledge Bases, and other AWS services into complete agentic AI applications.

Amazon Bedrock AgentCore

Deploy and operate AI agents securely at scale - using any framework and model.



Transcript 

AI agents hold immense potential,

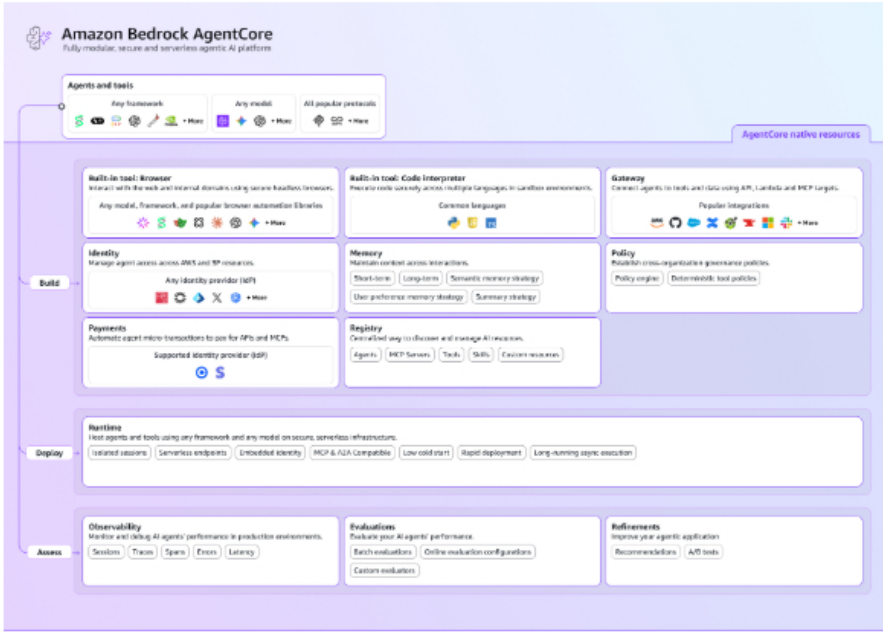
but today many of them are stuck in prototype purgatory.

Building agents is the easy part.

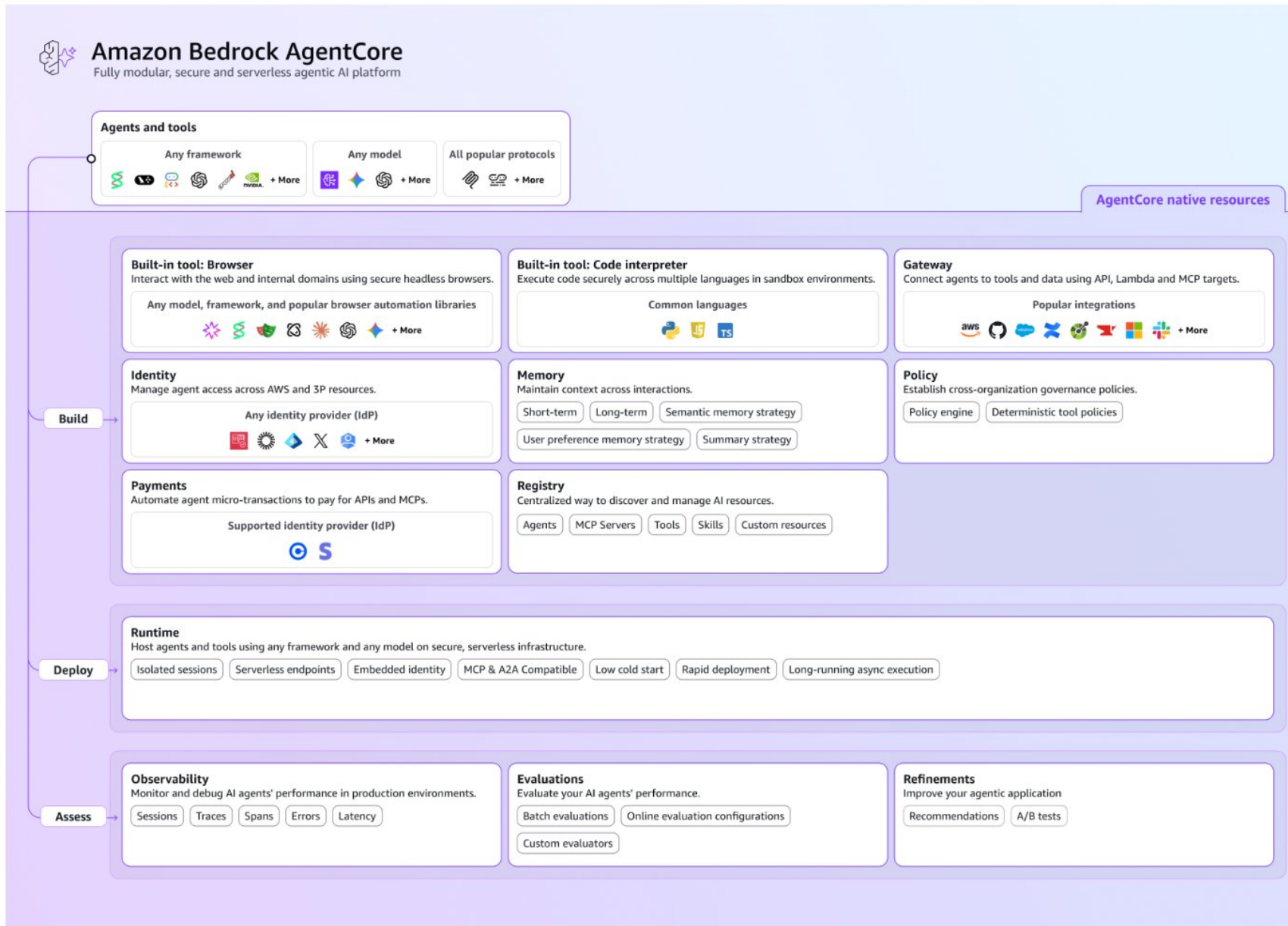
The real challenge is taking them from prototype to production to bring their AI agents to life at scale,

developers need specialized infrastructure and controls that don't exist off the shelf and require significant undifferentiated heavy lifting to stand up.

Amazon Bedrock AgentCore capabilities



Amazon Bedrock AgentCore capabilities



AgentCore is modular: you can use these capabilities together or independently, with any framework and any foundation model. The capabilities below are the ones you will position most often with customers.

- ▶ Runtime
- ▶ Memory
- ▶ Identity
- ▶ Gateway
- ▶ Code Interpreter
- ▶ Browser Tool
- ▶ Observability
- ▶ Policy
- ▶ Evaluations
- ▶ Harness
- ▶ Registry
- ▶ Optimization
- ▶ Payments

Common use cases for Amazon Bedrock AgentCore

1. **Equip agents with built-in tools and capabilities:** Leverage built-in tools (browser automation and code interpretation) in your agent. Enable agents to seamlessly integrate with internal and external tools and resources. Create agents that can remember interactions with your agent users.
2. **Deploy securely at scale:** Securely deploy and scale dynamic AI agents and tools, regardless of framework, protocol, or model choice without managing any underlying resources with seamless agent identity and access management.
3. **Test and monitor agents:** Gain deep operational insights with real-time visibility into agents' usage and operational metrics such as token usage, latency, session duration, and error rates.

Infrastructure



Amazon SageMaker AI

Bringing together widely adopted AWS machine learning (ML) and analytics capabilities, the next generation of Amazon SageMaker AI delivers an integrated experience for analytics and AI with unified access to all your data. Collaborate and build faster from a unified studio using familiar AWS tools for model development in SageMaker AI, generative AI, data processing, and SQL analytics, accelerated by Kiro. Access all your data whether it's stored in data lakes, data warehouses, or third-party or federated data sources, with governance built in to meet enterprise security needs.

- **AI and ML:** Build, train, and deploy ML models - including FMs - for any use case with fully managed infrastructure, tools, and workflows.
- **Unified Studio:** Build with all your data and tools for analytics and AI in a single development environment.
- **Catalog:** Securely discover, govern, and collaborate on data and AI with SageMaker Catalog, built on Amazon DataZone.
- **Lakehouse architecture:** Unify data access across Amazon S3 data lakes, Amazon Redshift data warehouses, and third-party and federated data sources with a lakehouse architecture.

AI compute

- ▶ Serverless
- ▶ Container-based
- ▶ EC2 Instances
- ▶ **Storage & Data**

Customer Path Selection

Customer approaches

 The future is built on agents and tools that work together.



There are three common customer approaches for implementing AI agents:

1. Implementing ready-to-use solutions
2. Building custom agents
3. Leveraging AWS Partners for specialized expertise

Safety Considerations: All paths include monitoring, human oversight options, and gradual autonomy increase.

1 - Implementing ready-to-use solutions

- ▶ Services
- ▶ Reliability
- ▶ Key benefits
- ▶ Implementation path
- ▶ Best for
- ▶ Considerations
- ▶ Implementation guidelines
- ▶ Recommendations
- ▶ When to recommend?

2 - Building custom agents

- ▶ Services
- ▶ Reliability
- ▶ Key benefits
- ▶ Development framework
- ▶ Best for
- ▶ Considerations
- ▶ Implementation guidelines
- ▶ Recommendations
- ▶ When to recommend?

3 - Leveraging AWS Partners for specialized expertise

- ▶ Services
- ▶ Reliability
- ▶ Key benefits
- ▶ Partnership model
- ▶ Best for
- ▶ Considerations
- ▶ Implementation guidelines
- ▶ Recommendations
- ▶ When to recommend?

Comparison matrix:

Features	1- Ready-to-use	2- Custom	3 - Partner
Time to market	Fast	Slow	Medium
Customization	Low	High	Medium
Initial cost	Low	High	Medium
Ongoing cost	High	Medium	Low
Risk level	Low	Medium	Low
Control	Low	High	Medium
Expertise Needed	Low	High	Medium
Maintenance	Low	High	Medium

The Partner-led approach offers organizations an optimal balance between rapid deployment and customization, avoiding the limitations of ready-to-use solutions while eliminating the complexities of building custom systems from scratch. By choosing to work with AWS Partners, customers benefit from proven expertise and established frameworks, reducing initial costs and implementation risks.

AWS Partners bring industry-specific knowledge and best practices, enabling medium to high levels of customization without the extensive time and resource investments required for fully custom solutions. This approach also provides sustainable long-term value through lower ongoing costs and shared maintenance responsibilities. To capitalize on this opportunity, AWS Partners should focus on developing deep agentic AI expertise, building repeatable solution frameworks, and maintaining strong AWS technical capabilities to deliver this balanced value proposition effectively.

Knowledge check



1. A large healthcare organization needs to implement AI agents for patient data processing within 3 months. They have limited internal AI expertise but require strict compliance with healthcare regulations. Which approach would be most appropriate?

Incorrect.	Incorrect.	Correct. Given the healthcare-specific compliance requirements, tight timeline, and limited internal AI expertise, getting	Incorrect.
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2. A tech startup wants to create a unique AI-powered customer service solution with proprietary algorithms and complete control over the feature roadmap. They have a strong technical team and are willing to accept moderate risk. Which approach should they take?

Incorrect.	Correct. The requirement for proprietary algorithms, complete control over features, and the presence of a strong technical	Incorrect.	Incorrect.
-------------------	--	-------------------	-------------------

3. A retail company needs to quickly automate standard customer service processes. They have limited budget for custom development, require proven reliability, and want minimal risk. Which approach would best suit their needs?

Incorrect.	Incorrect.	Correct. Given the requirements for quick implementation, proven reliability, limited budget, and minimal risk tolerance, turnkey	Incorrect.
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▼ Question 1

Multiple Choice

Time to answer: 52 seconds

Answer status:  Correct

Question

What is context persistence in relation to AI agents?

Answer options

Option	Correct answer	Your selection
A. The ability to forget all previous interactions		Not selected
B. Limiting interactions to a single context		Not selected
C. Resetting the agent's memory after each task		Not selected
D. Maintaining and learning from past interactions to improve future outcomes	 Correct	Selected

▼ Question 2

Multiple Choice

Time to answer: 18 seconds

Answer status:  Correct

Question

What are the three main value categories for AI agents in enterprises?

Answer options

Option	Correct answer	Your selection
A. Data analysis, customer service, and product development		Not selected
B. Marketing, sales, and operations		Not selected
C. Workplace productivity, business workflow, and innovation and research	 Correct	Selected
D. Cost reduction, revenue generation, and customer satisfaction		Not selected

▼ Question 3

Multiple Choice

Time to answer: 15 seconds

Answer status:  Correct

Question

According to the Agentic AI Maturity Scale, what is the highest level of agency?

Answer options

Option	Correct answer	Your selection
A. Generative AI Assistants		Not selected
B. Fully autonomous agentic systems	 Correct	Selected
C. Goal-Driven AI agents		Not selected
D. Rule-Based RPA		Not selected

▼ Question 4

Multiple Choice

Time to answer: 39 seconds

Answer status:  Correct

Question

Which approach is recommended for sales professionals when identifying customers interested in AI automation?

Answer options

Option	Correct answer	Your selection
A. Avoiding discussions about potential risks		Not selected
B. Proposing solutions without understanding specific use cases		Not selected
C. Addressing safety concerns early in the conversation	 Correct	Selected
D. Focusing primarily on technical specifications		Not selected

▼ Question 5

Multiple Choice

Time to answer: 15 seconds

Answer status:  Correct

Question

What is the primary challenge that agentic AI addresses, according to the market opportunity section?

Answer options

Option	Correct answer	Your selection
A. The shift from isolated AI tools to interconnected, autonomous systems	 Correct	Selected
B. The need for better data storage solutions		Not selected
C. The high cost of traditional software licenses		Not selected
D. The lack of skilled AI developers in the market		Not selected

▼ Question 6

Multiple Choice

Time to answer: 13 seconds

Answer status:  Correct

Question

How do AWS Partners with Generative AI Competency impact customer implementation success?

Answer options

Option	Correct answer	Your selection
A. They lower operational costs by 30%		Not selected
B. They increase production deployment rates by 30%	 Correct	Selected
C. They reduce implementation time by 30%		Not selected
D. They improve system performance by 30%		Not selected

▼ Question 7

Multiple Choice

Time to answer: 14 seconds

Answer status:  Correct

Question

What is a key characteristic of agentic AI?

Answer options

Option	Correct answer	Your selection
A. It requires constant human supervision		Not selected
B. It is limited to text-based interactions		Not selected
C. It demonstrates true agency and autonomous behavior	 Correct	Selected
D. It only works with structured data		Not selected

▼ Question 8

Multiple Choice

Time to answer: 15 seconds

Answer status:  Correct

Question

What is the primary benefit of using AWS Marketplace for AI agent solutions?

Answer options

Option	Correct answer	Your selection
A. It provides free AI agents for all users		Not selected
B. It offers unlimited storage for AI models		Not selected
C. It streamlines procurement and deployment while providing security and control	 Correct	Selected
D. It replaces the need for in-house developers		Not selected

▼ Question 9

Multiple Choice

Time to answer: 16 seconds

Answer status:  Correct

Question

Which AWS service is described as a comprehensive, secure, and flexible service for building generative AI applications and agents?

Answer options

Option	Correct answer	Your selection
A. Amazon RDS		Not selected
B. Amazon EC2		Not selected
C. Amazon SageMaker AI		Not selected
D. Amazon Bedrock	 Correct	Selected

▼ Question 10

Multiple Choice

Time to answer: 14 seconds

Answer status:  Correct

Question

Which value proposition does agentic AI deliver to organizations according to the course?

Answer options

Option	Correct answer	Your selection
A. Reduced need for employee training		Not selected
B. Autonomous process orchestration and decision-making	 Correct	Selected
C. Simplified data center management		Not selected
D. Lower hardware costs for computing		Not selected

▼ Question 11

Multiple Choice

Time to answer: 19 seconds

Answer status:  Correct

Question

Which of the following is described as a key enabler for transforming agentic AI from hype to reality?

Answer options

Option	Correct answer	Your selection
A. Quantum computing infrastructure		Not selected
B. Advanced model reasoning capabilities	 Correct	Selected
C. Blockchain technology		Not selected
D. Virtual reality integration		Not selected

▼ Question 12

Multiple Choice

Time to answer: 18 seconds

Answer status:  Correct

Question

How does the new API-based delivery model launched by AWS benefit organizations?

Answer options

Option	Correct answer	Your selection
A. It simplifies access to partner solutions through API calls	 Correct	Selected
B. It replaces all existing software with AI agents		Not selected
C. It enables direct access to AWS internal systems		Not selected
D. It provides free AI services to all customers		Not selected

▼ Question 13

Multiple Choice

Time to answer: 22 seconds

Answer status:  Correct

Question

What does the course suggest as a key capability that technical teams need to develop for AI agent systems?

Answer options

Option	Correct answer	Your selection
A. Cloud infrastructure management		Not selected
B. Advanced database administration		Not selected
C. Understanding the unique architecture of agent-based systems	 Correct	Selected
D. Basic programming skills		Not selected

▼ Question 14

Multiple Choice

Time to answer: 21 seconds

Answer status:  Correct

Question

What is the primary advantage of AWS Generative AI Competency Partners?

Answer options

Option	Correct answer	Your selection
A. Their expertise and success in implementing generative AI solutions has been validated by AWS	 Correct	Selected
B. They can bypass all AWS security protocols		Not selected
C. They have exclusive access to AWS resources		Not selected
D. They offer the lowest prices for AI services		Not selected

▼ Question 15

Multiple Choice

Time to answer: 14 seconds

Answer status:  Correct

Question

What is a key feature of Amazon Bedrock in relation to AI development?

Answer options

Option	Correct answer	Your selection
A. It offers access to various foundation models from different providers	 Correct	Selected
B. It replaces the need for any custom coding in AI development		Not selected
C. It automatically generates production-ready AI applications		Not selected
D. It provides unlimited free storage for AI models		Not selected

▼ Question 16

Multiple Choice

Time to answer: 14 seconds

Answer status:  Correct

Question

What is the primary difference between generative AI and agentic AI in decision-making?

Answer options

Option	Correct answer	Your selection
A. There is no significant difference in decision-making capabilities		Not selected
B. Generative AI makes all decisions autonomously		Not selected
C. Agentic AI requires constant human input		Not selected
D. Generative AI provides suggestions, while agentic AI makes autonomous choices	 Correct	Selected

▼ Question 17

Multiple Choice

Time to answer: 18 seconds

Answer status:  Correct

Question

Which capability of Amazon Bedrock AgentCore allows AI agents to remember past interactions?

Answer options

Option	Correct answer	Your selection
A. Runtime		Not selected
B. Gateway		Not selected
C. Identity		Not selected
D. Memory	 Correct	Selected

▼ Question 18

Multiple Choice

Time to answer: 53 seconds

Answer status:  Correct

Question

What is one of the key capabilities that Amazon Bedrock offers for AI development?

Answer options

Option	Correct answer	Your selection
A. Data customization tools to tailor AI to specific businesses	 Correct	Selected
B. Free unlimited cloud storage for AI models		Not selected
C. Automatic code generation for all AI applications		Not selected
D. Direct integration with social media platforms		Not selected

▼ Question 19

Multiple Choice

Time to answer: 15 seconds

Answer status:  Correct

Question

What is the primary purpose of Amazon Bedrock AgentCore?

Answer options

Option	Correct answer	Your selection
A. To manage human resources in AI companies		Not selected
B. To provide basic coding tutorials		Not selected
C. To offer cloud storage solutions		Not selected
D. To deploy and operate AI agents securely at scale	 Correct	Selected

▼ Question 20

Multiple Choice

Time to answer: 3 minutes 26 seconds

Answer status:  Correct

Question

What does AWS envision as the future of AI agents?

Answer options

Option	Correct answer	Your selection
A. Being limited to simple, single-step tasks		Not selected
B. Replacing human workers entirely		Not selected
C. Operating only in consumer settings		Not selected
D. Working alongside humans in various settings	 Correct	Selected